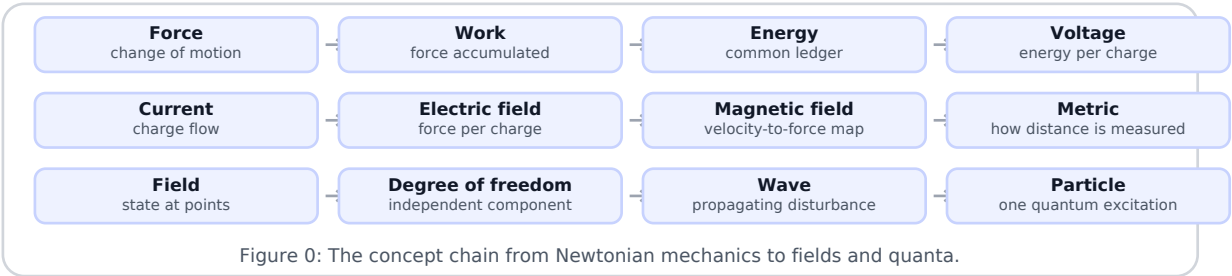


Physics Origins 02: From Work and Voltage to Fields, Waves, and Quanta

Logical Version: connecting the post-Newtonian concept chain into one line

V5 Logical / EN / 2026-06-30

This version keeps one main thread: how mechanics builds the work-energy ledger; how electricity defines charge, field, voltage, current, and resistance; how fields and degrees of freedom unify waves; and why quantization turns wave modes into countable particles.



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1. From force to work and the energy ledger

Newtonian mechanics first solved the question: why does motion change? Force is not merely the sensation of pushing or pulling; it is the cause of momentum change. This level answers a time-direction question: how does momentum change with time?

Core form of force: $F = dp/dt$

But force alone is not enough. The same force applied through 1 meter and through 100 meters produces different effects. Later physics therefore needed to accumulate force along displacement. That accumulated quantity is work.

Work: $W = \text{integral } F \, dx$. For a straight displacement in the same direction: $W = F s$

Work then forces the concept of energy. Work does not disappear; it can become kinetic energy, potential energy, heat, deformation, sound, and other forms. Energy becomes the common ledger: different effects can convert into each other, while the total ledger is constrained by conservation.

Concept	Core question	Logical role
Force	Why does motion change?	Rate of change of momentum over time
Work	How much effect is accumulated along displacement?	Force accumulated over space
Energy	How can different effects be kept in one ledger?	Transferable, storable, convertible ledger
Power	How fast is energy transferred?	Rate of change of energy over time

Conclusion: force explains change of motion; work explains spatial accumulation of force; energy puts many effects into one accounting system.

2. Charge, electric fields, and Coulomb law: localizing interaction

Electricity also begins from measurable phenomena: charged bodies attract or repel. Coulomb used a torsion balance to measure force, then varied distance and relative charge amount. The result was that electric force decreases with the square of distance and is proportional to the product of the two charges.

Coulomb law: $F = k q_1 q_2 / r^2$

This resembles Newtonian gravity not because mass and charge are the same thing, but because both are long-range interactions in three-dimensional space. A point source spreads its influence over a sphere; the sphere area grows as r^2 , so a $1/r^2$ law naturally appears.

The next move is conceptual. Physics rewrites "the force between two charges" as "a local force rule at every point in space." The source charge establishes an electric field; a test charge at a point feels a force there. In this way the interaction is localized.

Electric field: $E = F/q$, so $F = q E$

Quantity	Intuitive meaning	How it is defined
Charge q	Source and object of electromagnetic interaction	Compared through attraction, repulsion, and charge sharing
Electric field E	The local force rule for a unit charge	Electric force per unit charge
Coulomb force F	Interaction between two charges	Determined by distance and charge amounts

Conclusion: the electric field is not a mystical extra object; it is the local rule for force on charge.

3. Voltage, current, and resistance: circuit quantities from energy and charge

Current is the most direct circuit quantity. It is not the change of total charge inside an object; it is the amount of charge passing through a cross-section per unit time.

Current: $I = dQ/dt$. Unit: $A = C/s$

Voltage is not the amount of charge and not the amount of current. Voltage is energy difference per unit charge. In other words: how much energy one unit of charge can gain or release between two positions.

Voltage: $V = dE/dQ$. Unit: $1 V = 1 J/C$

Resistance describes how difficult it is for charge to flow. Under the same voltage, different materials allow different currents. That proportional relation is compressed into resistance.

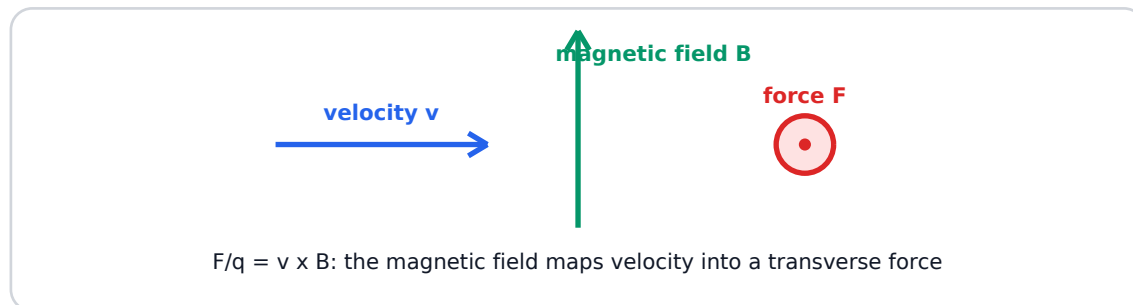
Quantity	Defining intuition	Form
Charge Q	How much charge exists	Q
Current I	How fast charge flows	dQ/dt
Voltage V	How much energy difference per unit charge	dE/dQ
Power P	How fast energy is transferred	$dE/dt = V I$
Resistance R	How hard flow is	V/I

Conclusion: voltage means energy difference per unit charge; current means charge flow rate; resistance means opposition to flow.

4. Magnetic field: a map from velocity to transverse force

The electric field is direct: a unit charge placed at a point feels a force. The magnetic field cannot be defined this way, because a stationary charge in a pure magnetic field feels no magnetic force. Magnetism appears as force only when charge is moving.

Pure magnetic part: $F/q = \mathbf{v} \times \mathbf{B}$



Therefore the magnetic field is not a force along the magnetic-field direction. It is better understood as a mapping rule: given the velocity of a moving charge, the magnetic field determines the transverse force it feels. The cross product encodes the direction relation: magnetic force is perpendicular to both velocity and magnetic field.

Field	Test object	Definition logic
Electric field E	Unit charge	Place it there; it feels force
Magnetic field B	Moving unit charge	Move through it; it feels transverse force

Conclusion: electric field is force per unit charge; magnetic field is a mapping from velocity of charge to transverse force.

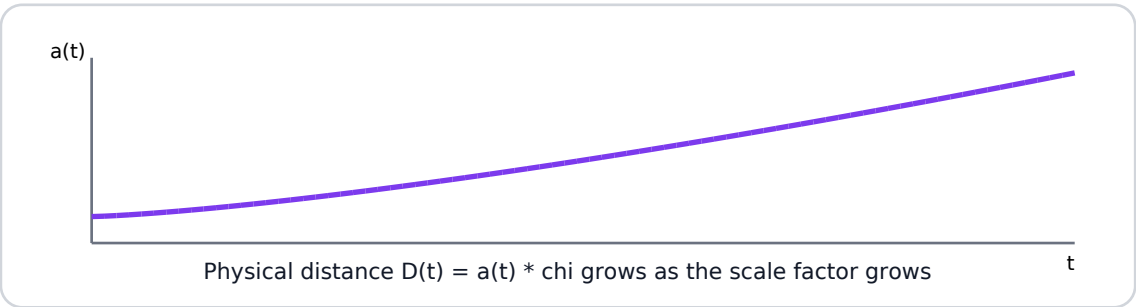
5. Units, measures, and metrics: from how much to how to measure

A physical quantity first has to be measurable. Length, mass, charge, and energy can be understood as "unit plus measure": first decide what counts as one, then state how many ones there are. Velocity, current, voltage, and power are one level higher: ratios or rates of change.

Level	Question answered	Example
Unit	What counts as one?	1 m, 1 C, 1 J
Measure	How many ones?	$L = 3\text{ m}$, $Q = 2\text{ C}$
Ratio / rate	How much per unit, or how fast?	$v = dx/dt$, $I = dQ/dt$, $V = dE/dQ$
Metric	How are distance or spacetime interval computed?	$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$

A metric is not merely a unit. It specifies how distance, time interval, and causal structure are computed. In Euclidean geometry the metric is fixed and flat. In general relativity the spacetime metric itself can vary. Cosmic expansion is a typical example: distant galaxies are not simply flying faster than light through fixed space; rather, the physical distance corresponding to the same comoving coordinate separation grows as the scale factor grows.

Cosmic expansion: $D(t) = a(t) \chi$



Conclusion: measure answers "how much"; metric answers "how to measure." In relativity, the metric itself is a dynamical physical object.

6. Fields and degrees of freedom: what is placed at each point

A field can be understood as follows: at each point, place a set of state variables. The state value may be a number, a vector, a tensor, or a more abstract object.

Abstract field picture: point -> state value



A field assigns state variables to points; degrees of freedom are the independent components

A degree of freedom is not the name of a thing. It is a variable that can change independently and must be specified to describe the system state. Water is a system; surface height, velocity, pressure, and density are degrees of freedom. The electromagnetic field is a system; the components of E and B are degrees of freedom. Spacetime geometry is a system; metric components are degrees of freedom.

System	Degrees of freedom / state variables	Associated disturbance
Water / fluid	surface height, velocity, pressure, density	water waves, sound waves, surface waves
Electromagnetic field	electric field, magnetic field, or potential	electromagnetic wave
Spacetime geometry	metric components	gravitational wave
Quantum field	field modes and excitation number	elementary particles or quasiparticles

Conclusion: a field is the structure that carries state variables; degrees of freedom are the independent components of those variables.

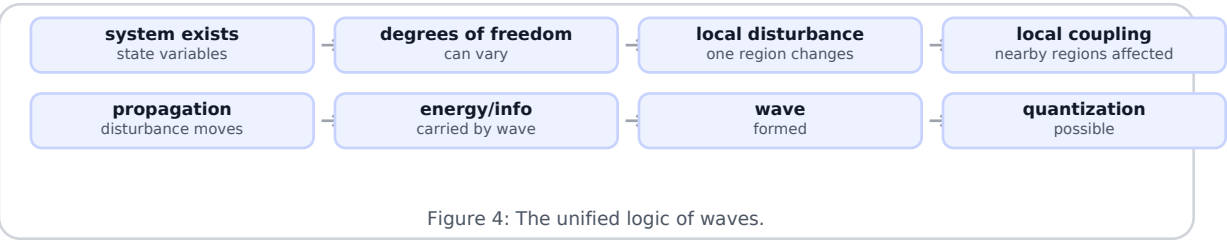
7. A unified definition of waves: local disturbance of system degrees of freedom

A wave should not be defined as "mechanical medium vibration." A mechanical medium is only one implementation of waves. A more unified definition is this: a physical system has spatially distributed degrees of freedom; those degrees of freedom are locally disturbed; local coupling propagates the disturbance outward.

Wave = local disturbance of a physical system's degrees of freedom, propagated outward.

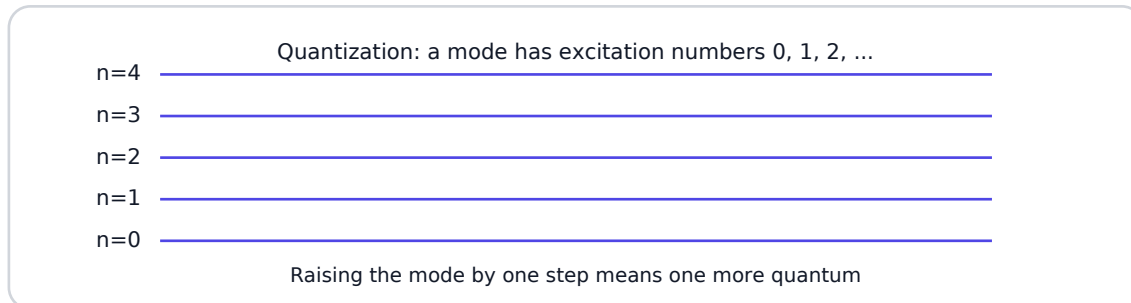
Wave	What must exist first?	What degree of freedom is disturbed?
Water wave	water / fluid system	surface height, velocity, pressure
Sound wave	air, solid, or liquid	density, pressure, velocity field
Electromagnetic wave	electromagnetic field	electric and magnetic fields
Gravitational wave	spacetime geometry	metric

This preserves both unity and difference. A water wave needs water; an electromagnetic wave needs the electromagnetic field. Neither is an object as a whole flying through space. Both are propagation of system state, energy, and information.



8. Quantization and wave-particle duality: wave is propagation, particle is counting

A quantum is not one particular particle. It is the concept of "one unit." After a field or a wave mode is quantized, its excitation number is not an arbitrary continuous value. It becomes countable: 0, 1, 2, 3, and so on.



A "discrete excitation" is not itself a physical quantity with SI units. It is more like increasing the particle number by one. This count has no unit. The excitation, however, carries attributes such as energy, momentum, charge, and spin.

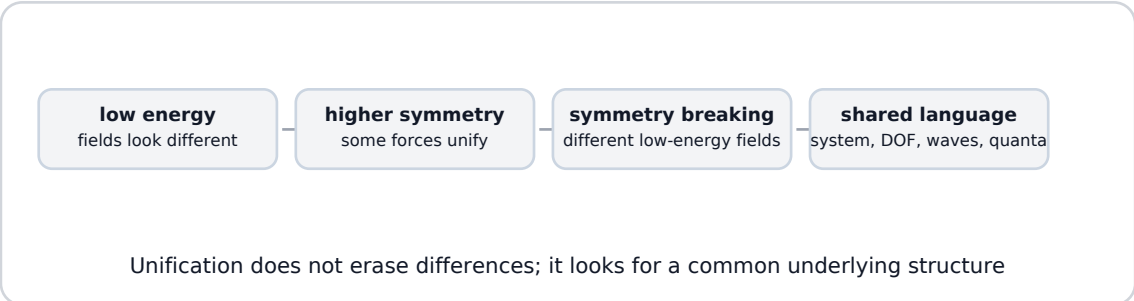
Object	Unified description	Example
Photon	one excitation of the electromagnetic field	discrete excitation of quantized electromagnetic wave
Electron	one excitation of the electron field	elementary particle
Phonon	one excitation of collective vibration in matter or fluid	quantized sound or lattice vibration
Ripplon	one excitation of a liquid surface mode	quantized surface wave

Wave-particle duality is then clearer. During propagation, the relevant object is the field state or probability amplitude; it can superpose, interfere, and diffract. During detection, the relevant object is excitation number and energy exchange; it appears in countable units.

Conclusion: wave is how something propagates; particle is how it is counted and exchanged. The same quantum object propagates like a wave and is detected like a particle.

9. Unified field theory: different appearances of one larger structure

Unified field theory does not mean that all fields are literally identical, nor does it force everything into one medium. Its goal is to find a deeper mathematical structure in which different fields and different interactions appear as different manifestations under different conditions.



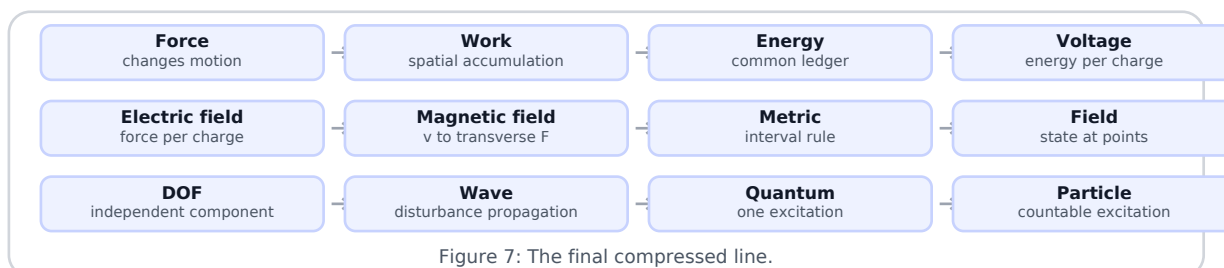
Level	What is unified	Status
Electroweak unification	electromagnetic and weak interactions	mature theory, strongly verified
Grand unified theory	strong, weak, electromagnetic interactions	theoretical candidates
Quantum gravity	gravity and quantum fields	unfinished
String theory / emergent spacetime	particles, fields, and geometry may arise from deeper structure	frontier theoretical direction

The value of this language is that it does not require water waves, electromagnetic waves, and gravitational waves to be "the same thing." It places them in the same modeling chain: system - degree of freedom - disturbance - propagation - quantization - discrete excitation.

10. Compressed conclusion

The post-Newtonian development of concepts can be read as one continuous chain. First comes force, explaining change of motion. Then come work and energy, explaining how effects accumulated along space can be accounted for in one ledger. Then electricity defines charge, electric field, voltage, current, and resistance as measurable relations.

Physics then abstracts "the state at a location" into fields, and calls the independently variable components of a system its degrees of freedom. A wave is not an object moving as a whole; it is a propagating disturbance of degrees of freedom. After quantization, the excitation number of a wave mode becomes discrete, and particles appear as countable excitations.



Final sentence: the physical world can be described through how a system's degrees of freedom change, propagate, and become quantized; waves are propagation patterns, particles are discrete counts, and fields are structures carrying state variables.

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